

- Q.29 Explain Heat Insulating materials.
 Q.30 Explain the effect of various alloying elements in steel.
 Q.31 Write short note on face centred cubic.
 Q.32 Define plastic deformation.
 Q.33 state the properties and uses of white metals.
 Q.34 What are the properties and uses of copper?
 Q.35 What are the various properties of ceramics?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Define heat treatment process. List the various heat treatment processes any explain any two.
 Q.37 Draw and explain iron-carbon diagram.
 Q.38 Explain the various point defects in metals.

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2nd Sem / Mech. Engg. (MSIL) Sub : Materials & Metallurgy

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The property of a metal by virtue of which it is able to bear the shocks is called
 a) Toughness b) Brittleness
 c) Hardness d) None of above
- Q.2 Atomic packing factor for simple cubic structure is
 a) 0.52 b) 0.48
 c) 0.74 d) None of above
- Q.3 Fatigue results in
 a) Elongation b) Ductile fracture
 c) Brittle fracture d) Crack formation
- Q.4 Cast iron has carbon percentage varying between
 a) 1 to 2 b) 2 to 4
 c) 4 to 6 d) 6 to 8
- Q.5 A liquid phase produces two solid phases during _____ reaction upon cooling
 a) Eutectic b) Eutectoid
 c) Peritectic d) Peritectoid

Q.6 The rapid cooling of component in water or oil or brine solution is called

- a) Hardening b) Quenching
- c) Austemering d) None of these

Q.7 Bakelite is the trade name of

- a) PVC b) Phenolics
- c) Polyesters d) None of these

Q.8 Polythene and PVC are manufactured by the process of

- a) Addition condensation
- b) Co-Polymerization
- c) Condensation polymerization
- d) None of these

Q.9 In concrete binding material is

- a) Sand b) Cement
- c) Stone d) None of these

Q.10 Aluminium oxide is produced from

- a) Cryolite b) Bauxite
- c) Duralumin d) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define metals.

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Q.12 The percentage of carbon in medium carbon steel varies from _____ to _____ %.

Q.13 Define bio materials.

Q.14 Define unit cell.

Q.15 What is body centred cubic structure.

Q.16 Define ductility.

Q.17 The trade name of thermocole is _____.

Q.18 Name any two surface defects.

Q.19 P.V.C. becomes soft beyond _____ temperature.

Q.20 Conductivity of metal is more than conductivity of non-metal. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Define any five mechanical properties of material.

Q.22 Give the properties and its uses of stainless steel.

Q.23 Define the term twinning with neat diagram.

Q.24 What is plastic. Write its application in automobile sector.

Q.25 Draw the basic process of making iron and steel.

Q.26 What are the various allotropic forms of iron?

Q.27 Give some properties and uses of asbestos.

Q.28 How will you classify materials?

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